Requirement from Ethiopia

Special Formal Tender for the Supply of a Credit Reference Databank at the National Bank of Ethiopia

1. CREDIT REGISTRY SYSTEM

A credit registry is a unique database of credit information (credit history) on persons natural or corporate within a particular country or region used to evaluate and ascertain credit risk. Credit registries are generally developed to aid prudential supervision of financial institutions It is a pillar of sound banking practice used in customer due diligence as a tenet of prudent know your customer practice.

Overview of the Credit Reference System

1.1. The application is a web-based solution that is designed and developed by xxxx The system is customisable to meet the requirements of the National Bank of Ethiopia. This solution will serve the Central Bank of Ethiopia and will cater for all Commercial Banks and microfinance Units and SACCOS. It has expandable function to serve the non-regulated units in retailing, agriculture manufacturing, mining and education. The solution will be accessible through the internet and will be able to receive, validate, and merge credit information from banks, financial institutions, and other institutions, based on the requirements of the central bank.

1.2. The Credit Registry will incorporate personal information, domicilia, education and employment, loans and loan performances, past applications, company registrations and directorships, bankruptcies, criminal and civil judgements as well as politically exposed persona information. This is to aid in proper risk profiling and in some jurisdictions risk pricing. The Registry are mainly used by credit yet they are useful to all sectors of the economy including the citizens in order to avoid over indebtedness and cultivate a culture of responsible bank use practice

1.3. We intend to develop for the Bank of Ethiopia a Credit Registry that will be user-friendly for policy and regulatory purposes, such as monitoring the credit risk of the financial system and conducting macro prudential oversight whilst promoting responsibility bank use. This system will be customised to the needs of Bank and will work with other virtual environments e.g. Microsoft 365 and emails.

1.4. Credit registry offer information on an individual about:

- Credit agreements, both past and present
- How much you owe and your repayment history.
- Bankruptcies and insolvencies on your record
- Court judgements
- Political exposed status

1.5. Banks offer extra information in their credit references, including the type of account someone has and how long they've had it for, as well as details of any late payments or overdrafts.

1.6. There is growing calls for developing nations to develop Credit registries as they are becoming increasingly important for several reasons:

1.6.1. **Improved access to credit**: Credit registries can help to improve access to credit for borrowers by making it easier for lenders to assess their creditworthiness. This can be especially beneficial for small and medium-sized enterprises (SMEs), which often have difficulty obtaining financing from traditional banks.

1.6.2. **Reduced risk of financial instability**: Credit registries can help to reduce the risk of financial instability by making it more difficult for borrowers to take on excessive debt. This is because lenders can use the information in the registry to track borrowers' borrowing history and identify those who are at risk of default.

1.6.3. **Greater financial inclusion**: Credit registries can help to promote financial inclusion by making it easier for people who have been excluded from the formal financial system to access credit. This is because the information in the registry can be used to verify their identity and creditworthiness.

1.6.4. **Anti-money laundering and fighting terrorism tools.** Registries become increasingly important in monitoring individuals, wealth accumulation, and a useful tool in preventing funding for both domestic and international terrorism.

1.6.5. **Politically exposed persona reporting**

2. FUNCTIONALITY OF THE CREDIT REGISTRY SYSTEM

2.1. The software solution will be capable of accommodating borrower identity information, positive and negative data and all types of loan records in respect of commercial, consumer, SME and MFI credit transactions, including off balance sheet operations and the matching of all types of borrowers records from disparate regulated entities, including related party lending (company and related subsidiaries; individuals and related companies/businesses).

2.2. The solution will be interfacing to third party databases such as but not limited to licensed Credit Reference Bureaus, Core Banking Systems, Loans Management Systems, the Registrar’s office, and the Registrar of Companies.

2.3. The solution will have capacity to accept and support business intelligence tools and business reporting tools to be used by National Bank of Ethiopia in developing information for monitoring of the financial sector for systemic risk and oversight. The solution will have functionality to sort and differentiate bank account number in order to effectively distinguish contracts granted for each person.

2.4. The functionality of a credit registry system can be broken down into the following parts:

a. **Data sources**: Collection of data from various sources, including lenders (banks, microfinance institutions, etc.), public records, and other givers of credit.
b. **Data types**: The data collected includes personal information like name, address, and ID number, as well as details about credit accounts like loan amount, type, repayment history, and delinquencies. Also collecting company information like company name, operating address, and business registration number. Having the ability to edit and update searches.
c. **Data validation**: Before entering the registry, the data goes through a rigorous validation, which will also include maker-checker process to ensure accuracy, completeness, and consistency.
d. **API** (Application Programming Interface): This will allow machine to machine or system to system communication. This allows the registry system to send and receive data from multiple sources, for example other CRBs, Ministries, demographic data, courts, and other digital lending platforms.
e. Analytical tools for statistical analysis, demographic reporting, etc

2.5. Data Processing and searches

a. Batch processing online and offline
b. Individual applications on both natural and corporate persons
c. Portfolio monitoring
d. Data validation
e. Make use of drop down menu for accurate data processing

2.6. System tools and functions

a. Enables system admin to monitor activity
b. Scripts to manage system start-up, system shut down, maintenance, data back-ups, fail over to Disaster Recovery Site
c. Recovery tools to ensure seamless data recovery for all information in all database fields
d. Tools for performance monitoring processing and computer power consumption.
e. Document system management procedures
f. Scripts for system security
g. This system will be customised to the needs of the bank and will work with virtual environments e.g. Microsoft 365 and emails.
h. Data entry capabilities with short cut keys

2.7. **Process Flows**:

a. System to allow for users to run searches as per assigned roles
b. All searched persons to have granted consent for search to be conducted and system to be set in order to capture all tenets of such consent i.e. provision of robust KYC documentation.
c. All pending enquiries/searches to be escalated within defined times.
d. Work flow to cater for maker and checker functions and to include similar roles in API searches.
e. Reports generated to be viewed by user.
f. Admin reports to be viewed only by designated users.
g. Sensitive/intelligence based reports to be sent to designated individuals.
h. All reports to have time stamp, serial numbering, user credentials as well as a bar coding for extra layer of security on reports
i. System to allow admin to input alerts, reminders and other markers and be able to support such alerts.

3. USER INFORMATION, DATA STORAGE AND MAINTENANCE

3.1. The system will allow to register new users. The validated user information / data is stored securely in a central database, accessible only to authorized users to this data and can also be uploaded in batches. Availability of a Disaster Recovery site, which will allow recovering from major disruptions or disasters, by backup and recovery solutions, and mirroring, with the goal of restore critical operations as quickly as possible after a disaster, with acceptable data loss.

Send notification to users as changes made to the application through prompts, notifications, emails, sms or login pop up messages.

a. System to authenticate user through user ID, Password and Client's Network IP Address and/or OTP.
b. System to capture user information for audit trails.
c. System to have an extra security feature through PIN and generation of One Time Password (OTP.)
d. User information is stored in a secured and encrypted database.
e. Analytical tools for statistical analysis, demographic reporting, etc
f. System to alert admin user as a notification on a pending action item which may be accepted or rejected with reasons.

3.2. **Search functionality**: Ability for authorised users to search for records of individuals and companies, either by name, surname, identity number etc. Admin user has search function to include truncation, phrase searching, transactional, informational, navigational and nested searches.

3.3. **Data updates**: The ability of banks, microfinances and other financial institutions regularly updating information in the registry to reflect changes in individuals and companies credit situations. Processing of searches. This will also follow the maker checker principles, that's the updated record will have to be validated and approved.

3.4. **Data privacy**: Strict data privacy regulations and security measures are in place to protect borrower information from unauthorized access.

3.5. **Data uploads** and downloads through batch file submission and retrieval on the database supported by admin scripts to log time and task performed.

3.6. Disaster Recovery

a. System to have automated onsite and offsite DRP manual with DRP process to be adhered to and enforced.
b. Offsite DRP to include cloud solutions, network and power redundancy.
c. DRP at Bank should also include critical resource training and leadership training and development.
d. DRP solution to also include statistics of searches done to ensure no loss of revenue in disaster recovery.
e. Both manual and automated backups for DRP to be adhered to and enforced with the four eye principle being used.
f. Data archiving scripts for automated service with the archive being stored separately.

4. DATA ACCESS AND UTILIZATION:

4.1. **Credit reports**: Lenders can access credit reports on potential borrowers by submitting their identifying information. These reports provide a comprehensive overview of the borrower's credit history, helping lenders make informed lending decisions.

4.2. **Risk assessment**: Enabling lenders to assess borrowers' creditworthiness more accurately, reducing the risk of bad loans and promoting financial stability.

5. REPORTING

The credit registry will produce the following reports.

5.1. User activity reports.

- Access audit trails for additions to data
- Daily usage stats
- Number of searches done reporting
- Access level separation reporting

5.2. Credit reports.

- The most fundamental report, providing a comprehensive overview of an individual or business's credit history.
- Includes information on:
  - Personal information
  - Education and employment
  - Past applications / Credit inquiries
  - Company registrations and status
  - Directorship

  - Addresses used
  - Property ownership status
  - Loan performance
  - Outstanding debt balances
  - Public records (e.g., bankruptcies, liens, judgments)

5.3. Statistical reports.

5.4. Portfolio reports
- Aggregated views of credit risk within a lender's portfolio.
- Used to assess overall credit quality, identify trends, and manage risk exposure.
- Essential for portfolio management and regulatory compliance.

5.5. Collateral registry reports
- Collateral submitted
- Value of all Collateral
- Applications from the registry
- Accepted / granted facilities and value from the registry

5.6. Event triggered reports
- Set event as time specific, value specific or database size specified

5.7. Payment reports
- Credit reports generated for Subscribers
- Credit reports generated for the public
- Administration fees for record updates
- Credit scores generated of Credit Registry App
- Audit trail reports

5.8. Credit scores
- Notifications of significant changes in credit information, such as:
  - New credit applications
  - Changes in account status (e.g., late payments, defaults)
  - Identity theft or fraud suspicions

6. DATA SUBMISSION REPORTING

6.1. Logs of files submitted and unique filing in non-editable register.
6.2. Quality of data submitted reports
6.3. Quantum of records report
6.4. Automated service for reminders of late submission
6.5. Sectorial Analysis

7. JUDICIAL SERVICE REPORTS

7.1. Civil Judgement received and updated.
7.2. Summons issued pending trial.
7.3. Criminal judgements received and updated
7.4. Arrest brought before trial reports
7.5. Crime public media articles of merit

8. MET DEPARTMENT REPORTS.

8.1. Weather forecasts and index prediction of adverse weather damage

8.2. Rainfall pattern reports unique to provinces
8.3. Extreme weather Alerts

9. BUSINESS REQUIREMENTS TO BE MET BY THE SYSTEM

9.1. The system shall perform the following functions: receive, validate, and merge information from banks, financial institutions and from alternative data sources such as courts, business registries, tax offices.
9.2. The Registry will be customised to produce credit reports and other related products based on the available data for credit reference purposes and undertake detailed analysis of the data as well as provide data security and backup.

10. COMPUTING HARDWARE SPECIFICATIONS

10.1. Servers with will be supplied with enough computing power to allow for real time reporting for multiple concurrent users. The system will be customised to work optimally and cover primary, backup and provide for on-site and offsite disaster recovery environments.

11. Network and Communications Specifications

11.1. Local Area Network and Wide-Area Network
11.2. The application will make use of existing LAN and the system will allow all banks and other data providers to be able to provide and get data through the same network.
11.3. System will connect to peripheral hardware such as printers and scanners.

12. SOFTWARE SPECIFICATIONS

12.1. System Software and System-Management Utilities
12.2. The software will be able to use different fields present in the database to create a matching key to be used in matching data for identification of individuals or legal entities including related party lending (company and related subsidiaries; individuals and related companies/ businesses) such as combination of Identification Numbers, Date of Birth, Name, address and telephone numbers. The matching module shall be easily configurable and flexible in terms of setup of the set of matching rules
12.3. Specific reporting tools will be incorporated to be provide and generate ad hoc reports or extract data from the application database.

13. INTERFACES FOR DATA ACQUISITION

13.1. The software will be able to get data from data providers though various interfaces.
  a. Web-interface for manual data upload.
  b. Web-service for automatic upload of data.

c. Secure FTP protocol Solution will extract and / or prepare the data input file according to the desired input format as attached.

14. FEATURES

The Credit Registry System also comes with the following additional features which add more value to the application and make it more useful to clients.

14.1. **Registry System User**

The system supports the following user types:

a. Lenders / Users - These are regular user who has established an account with the registry to which fees for registration of notices and requests for certified search reports may be charged. These users will have access to credit reports,
   - Credit scores,
   - Portfolio monitoring,
   - Alert notifications
   - Data uploads

b. Administration users / Regulators – These are the users that provide system overview of the registry system. These will also provide day to day assistance to other users of the system. Any change to our profile will be sent to the user through email and/or sms.

14.2. Administrative Features

a. Membership administration
   i. Register of users
   ii. User credential set up
   iii. Access level set up maker and checker functionality
   iv. User status change registration showing reason

b. Application configuration
   - Set up reason for application in the registry
   - Count of application
   - Register of searches for invoicing purposes

15. PAYMENTS

15.1. Integrated Capability for Payment
15.2. Payment Audit Reporting Features
15.3. Manage Transaction Fee Schedule
15.4. Reconciliation of Payment Features

16. SYSTEM MANAGEMENT, ADMINISTRATION, AND SECURITY SPECIFICATIONS

16.1. Application capabilities

In addition to the management, administration, and security requirements specified in each section covering the various hardware and software components of the System, the System will have evidence best practice and also provide for the following management, administration, and security features at the overall system level.

17. TECHNICAL MANAGEMENT AND TROUBLESHOOTING

The software will provide for the following technical management and troubleshooting features:

17.1. Have explanatory error messages.
17.2. Have on-line tutorials for trouble shooting

18. USER AND USAGE ADMINISTRATION:

The software will provide for graphical user interfaces for end users and should be able to accommodate many concurrent users to access, submit and query information from the system with an acceptable real time response time of five (10) seconds when the application is tested within the infrastructure environment.

19. **SECURITY**:

19.1. The system will provide for the following extra security features to ensure full data integrity and confidentiality at all times.
   a. Limit user access to selected functions through user profiles/roles, that is, by module, function and screen;
   b. Provide strong user authentication mechanisms including two factor authentication features;
   c. Provide strong encryption mechanism to protect sensitive data that is stored/transmitted across the network;
   d. Configure user status to signal active, suspended, de-active, idle users.

19.2. The system standard user access security features will be:
   a. Passwords and User ID's
   b. Login Encryption / Validation
   c. Server Level User Authentication
   d. Tracking Mechanism of all Activity in System (Log History)
   e. Password Policy of Client stipulating length and complexity
   f. Access level differentiation and audit trails (separate interfaces for system administrators performing technical and administrative functions and users performing management functions in the system)
   g. Limit user access to selected functions through user profiles/roles
   h. Change of authentication credentials at predefined regular time intervals.
   i. Login attempt failures should be limited to three, and thereafter the system shall disable the user.
   j. Terminate / timeout idle sessions after user defined specified period.
   k. Configure of number of sessions by user profile

l. User profile password expiration

19.3. Audit Trails

a. Provide exact time full report of all logins/logouts to system.
b. The system's audit trail should tag all activities performed within the system with time stamp and user identity.
c. The system will have provision for recovering information in the event of disaster or total system failure.
d. Supervisory check (four-eye principle)
e. Audit trail reports are also role specific

19.4. The system will be capable of providing an interface for double control of important system changes as data load and data changes in the production database, changes in user profiles and administration. The software shall provide for graphical user interfaces for end users and should be able to accommodate many authorized concurrent users to access, submit and query information from the system with acceptable real time response time of five (10) seconds when the application is tested within the infrastructure environment

20. OTHER SERVICE SPECIFICATIONS

The system design/architecture will be done using varying development language, interface and integrate with other systems.

20.1. Data Migration

a. The application will be capable of handling data migration from existing data with minimal to no loss of data. The vendor will ensure seamless migration of data from legacy systems, current CRB as well as Banks and Microfinance and ensure minimum to no loss of data files.
b. Duplication will also be managed.
c. All processes of migrating data will be documented.
d. Where necessary licensing of data migration will be sort.

20.2. System Integration

The system may be integrated with an existing system within the user Bank. The system shall interface with Banks and Financial institutions for data submission; with Licensed Credit Bureaus and the Central Bank of Ethiopia Supervision Directorate for data utilization; and any other 3rd party databases such as Registrar's office and Registrar of Companies, for data validation.

20.3. Automation (API)

The system should be capable of providing necessary business to business (b2b) interfaces for 3rd party tools and users of the system. It will be able to interfere and integrate with other banks and subscriber systems.

20.4 Expandable Solution

The System will be expandable to include services such as skip trace report, Politically Exposed Persons and anti-money laundering reporting and other services required in the future.

21. TRAINING AND TRAINING MATERIALS:

21.1. For the user there shall be two types of users; data providers and data users:

a. Data providers shall be trained on data contribution requirements, databank operations, proper, timely filing, and submission of the regulated data elements to the databank, and the proper and timely resolution of disputes and the correction of any incorrect or incomplete data.
b. Data users shall be trained on data contribution requirements, databank operations and extraction of information from the databank and the proper and timely resolution of disputes.
c. Training manuals will be provided.

21.2. Technical staff shall be trained on the following:
a. Data contribution requirements, databank operations, proper filling, and submission of the data to the databank.
b. Data matching, sorting, and extraction.
c. System administration, Database administration, troubleshooting, backup, and recovery.

21.3. Technical Support: There shall be a warranty period of twelve (12) months for the software whereby the support shall be in the form of on-call basis or on-site depending on the severity of the incident.

21.4. User support / hot line: There shall be hot line available 12 hours for seven (7) days a week including public holidays.

21.5. Technical Assistance: There shall be a Technical and business experts on duty that will respond to technical issues within two (2) hours from the time incident is reported.

21.6. Post-warranty maintenance services: After expiration of the warranty period, the Supplier may at the Client’s request, provide software support on the terms and conditions as shall be agreed by the Parties in the contract.

22. USER ACCEPTANCE TESTING AND QUALITY ASSURANCE REQUIREMENTS

22.1. The system will allow for UAT prior commissioning. The UAT will be used to run test on application integrity, fit for purpose as well speed and quality of reports generation and score computation.

22.2. Provide a UAT environment for batch files and online searches for all portals.

23. LICENSING

23.1. Upon full implementation the vendor will pass all licensing for use to the National Bank of Ethiopia.
23.2. All source code will be surrendered.
23.3. All documentation outlining system architecture, system workflows and system reporting, troubling and database configurations will also be handed over.

24. POST IMPLEMENTATION

24.1. The Financial Clearing Bureau will maintain the system and provide updates, address system errors, expand system functions as requested by National Bank of Ethiopia. This will be covered in a separate agreement to the one under proposal.

REQUIREMENT FROM LIBERIA

B. Functional, Architectural and Performance Requirements

1.1 Legal and Regulatory Requirements to be met by the Information System

1.1.1 The Information System MUST comply with the following laws and regulations:

1.1.1.1 Liberia's Credit Reference Bureau Regulations # CBL/SD/003/2010 (CRB Regulation)

1.1.1.2 Any law and/or regulations governing data protection and technology transfer in Liberia

1.2 Business Function Requirements to be met by the Information System

1.2.1 The Information System MUST support the following business functions

1.2.1.1 The system will be expected to source and integrate data elements from non-bank institutions and public sources such as other government agencies (OGA) and third parties within the credit market in Liberia

1.2.1.2 The system must facilitate CBL's risk-based supervision of the financial sector

1.2.1.3 The system must have a fee and revenue management module that allows fee schedules to be defined for different types of data users, and integrates with national payment platforms to facilitate online payment via mobile money, bank transfers and card payments, generating electronic payment receipts in PDF and/or SMS text formats

1.2.1.4 The reporting functionality of the system must allow for the generation of the business user, regulatory/compliance and system performance reports on-demand or scheduled basis and shall allow for their dissemination using various means of communication including email and downloads in Excel or PDF formats

1.2.1.5 The system must automate at minimum, the following credit bureau functional requirements:

#	FUNCTIONAL REQUIREMENTS	PROPOSER'S RESPONSE		
		Complies	Needs Customization	Does Not Comply
CB1	The system must support both business and consumer credit data			

CB2	The system must be able to collect at minimum, the following **credit information**: amount of the loan or other facility granted to the creditor, sum of outstanding loans; date on which the loan was made, loan repayment schedule, details on collateral used to secure the debt obligation			
CB3	The system must be able to collect at minimum, the following details on the **individual borrower**: full name, gender, date of birth, place of residence, information contained in borrower's state-issued			
CB4	The system must be able to collect at minimum, the following details on the **Institutional borrower**: name of business, its organizational or legal form, location, date of registration, business's TIN, names, TIN and state-issued identity documents of Chief Executive Officer, board members, and beneficial			
CB5	The system must be able to accept records on liens made on property and data on court rulings			
CB6	The system must be able to collect and process borrower credit history from various sources including non-bank lenders, utility companies, telecom, courts and other public			
CB7	The system must be able to exchange and compile information from the collateral registry database			
CB8	The system must provide a mechanism to indicate or flag the existence of an ongoing dispute initiated by a consumer or relevant authority			
CB9	The system must facilitate standardized formats and templates as defined by the CBL for data			
CB10	For data feeds, the system must allow for minimum data sets to be defined for each category of data provider e.g. commercial banks, MFIs, VSLA, Mobile Network Operators, Utility companies, digital credit			
CB11	The system must be configurable to accept data in defined frequencies - daily, weekly, monthly			
CB12	The system must have capabilities to normalize, validate and verify each data load into its database e.g. validation routines to enhance data quality e.g. digit check, allowable range of values			
CB13	The system must be able to validate data subjects against the Liberia national identification database			

CB14	The system must automatically reject all submissions that fail to meet the CBL's data			
CB15	The system must have the facility to provide data validation result notification to data providers in real time and in a log file			
CB16	The system must be able to merge and analyze information collected from various sources to form a comprehensive credit history record for each borrower. records - The system should be able to map data collected from disparate sources on a data subject to the data subject. It should also be able to create, native to the system', such markers that uniquely identify data subjects on the system			
CB17	The system must provide history of borrower's payment performance with loan classification over time for 12 consecutive repayment periods			
CB18	The system must enforce who can receive credit information it holds (e.g. a credit report recipient as defined under CBL regulations, a recipient registered to obtain information)			
CB19	The system must provide full credit information history with both positive (on-time) and negative (delinquent) payment information on consumer and			
CB20	The system must be able to report on the status of loans and advances (i.e. restructured, number of times restructured, written off loan status etc.…)			
CB21	The system must be able to undertake bulk credit reference checks			
CB22	The system must allow for credit report formats to be customized for each category of data provider (i.e. bank and non-lenders etc.)			
CB23	The system must have functionality to maintain records of requests for the issue of credit reports and records of credit reports provided			
CB24	The system shall refuse to provide a credit report if the request for its provision contravenes a provision of CBL regulations or any other law.			
CB25	The system must be able to identify and report on all institutions that the potential borrower is indebted or exposed to			
CB26	The system must retain credit histories for a period of seven years or for a period as defined by CBL			

CB27	The system must have sophisticated matching algorithms and programs to derive the identity of a person based on various pieces of data			
CB28	The system must provide search functionality to search using one or more data fields e.g. borrower identifier, National ID, TIN, registration/loan reference number and other identifiers.			
CB29	The system must have the functionality to generate and print digitally signed search reports			
CB30	The system must assign a unique number identifying the search/inquiry result as a record on the system with time stamp.			
CB31	The system must allow data subjects to access and verify information stored about them			
CB32	The system must offer automated credit scoring functionalities. The credit scoring functionality must be able to use data that is past the retention period set by regulations			
CB33	The system must have an online complaint management module with a chatbot feature, alert functionalities and must be appropriately configured to provide online access to all stakeholders in the credit reporting complaint handling process (i.e. the CBL, data providers, data			
CB34	The system must have some data analytics functionality to support the compliance monitoring, research, statistical, and other policy needs of the CBL as well as other parties approved			
CB35	The system must be role-based and restrict access to modules and screens to users as defined by the CBL			

1.3 Architectural Requirements to be met by the Information System

1.3.1 The Information System MUST be supplied and configured to implement the following architecture.

1.3.1.2 Software Architecture:

a. The web-based user interface should require minimal or no installation of third-party software and must conform to WC3 standards
b. To minimize the total cost of operations for the system, it is recommended that the system should be deployed on opensource software
c. The system must provide a welcome page with explanatory notes on fees/ charges, specific help and support, terms and conditions of use, Helpdesk

contact information, standard features of login for users and retrieval/reset of forgotten user password and chatbot feature.

d. The user interface must support a minimum screen resolution for desktop devices of 1024x768 and avoid side-scrolling under any circumstances, and must be adaptive to mobile devices e.g. smart phones or tablet devices
e. The system shall provide the facility to save incomplete data entry processes for retrieval and later completion and submission by the user.
f. The system should be developed on microservices architecture and provide a full set of APIs for third-party system integrations for information exchange
g. The architecture should allow for new capabilities to be easily added or extended in the future
h. The system should be fault-tolerant and ensure high availability and recoverability in the event of a disaster
i. The system should enable easy troubleshooting, updates, and maintenance without affecting availability

1.3.1.2 Hardware Architecture:

a. The system must be capable of working in any type of server environment (on-premises and cloud) and print to any printer on any standard paper size.
b. Proprietary hardware components shall not be used for the end-user deployment of the system. All components of the deployment must conform to widely supported open standards.
c. The preferred server environment to deploy the system is a cloud environment optimized for efficient response times, throughput, and resource utilization to meet performance requirements even under peak loads.
d. The server environment must operate in a 3-tier server architecture be sized accordingly to scale up compute, memory, storage, and network capacity as needed to support business growth and changing demands.

1.4 Systems Administration and Management Functions Required to be met by the Information System

1.4.1 The Information System MUST provide for the following management, administration, and security features at the overall System level in an integrated fashion.

1.4.1.2 Installation, Configuration and Change Management:

a. The system must provide a facility to define all workflows for processing approvals
b. System must be able to detect if changes made by a user require approval by a privileged user and subsequently route the change for approval
c. The system shall support use of multiple currencies (at minimum, United Stated dollar-USD and Liberian dollar-LRD)

1.4.1.3 Operational Monitoring, Diagnostics, and Troubleshooting:

The software should provide for the following technical management and troubleshooting features:

a. The system must provide for explanatory error messages
b. The system must provide on-screen help facility

c. The system must provide a dashboard for system administration information to be displayed e.g. concurrent user connections and system resource utilization

1.4.1.4 <u>User Administration and Access Control</u>; User and Usage Monitoring and Audit Trails:

a. User access to the system must be role-based and the system should provide for a registered user to play one or multiple roles defined in the system
b. The system shall provide an on-line form for user registration. Submission of a user registration form will require formal acceptance of standard terms and conditions for use of the system.
c. The system shall validate each field in the online form as entered during the registration process and automatically identify errors and discrepancies for correction.
d. The system shall employ a unique identifier for each registered user
e. The system shall use screen pop ups to send alerts and notifications to logged on system users
f. The system must provide for a forensic audit trail of chronologically ordered events, timestamp assurance and tamper checks to ensure the integrity of database records

5. <u>System and Information Security and Security Policies</u>:

a. The system must comply with ISO 270001 standards and ensure 100% adherence to the Information Security Policy of the CBL
b. The system shall support use of two-factor authentication via email or SMS text messaging
c. The system shall ensure that all interfaces and information exchange with other systems are secured using the appropriate levels of encryption and authentication mechanisms.
d. The system must apply the maker/checker or four-eye principle to important system changes to data in the CR and CB databases and system and user profiles administration.
e. The system must work with common anti-virus software to ensure uninterrupted operation.
f. The system must provide for secure digital signatures of approving officers at the CBL
g. The system must provide an appropriate security mechanism for safeguarding changes to the embedded digital signatures of authorizing officers at the same time provision for changes in authorizing officers when required

1.4.1.6 <u>Back-up and Disaster-Recovery</u>:

a. The system shall conform to CBL's data retention and disaster recovery policies.
b. The system shall be configured for automatic rollback of the data backup on extreme disasters to the last point where data and system integrity was still retained.

c. The system shall be configured to create an archive of data on a prescribed frequency as part of its database clean-up and compression mechanisms which shall be easily accessible on demand.

1.5 Performance Requirements of the Information System

1.5.1 The Information System MUST reach the following performance levels.

1.5.1.1 The system must operate on low bandwidth footprint (at minimum 256 kbps) to ensure that remote uses can work at speeds that do not hamper their data input and submission

1.5.1.1 The system shall duly support configurations to handle up to 100 concurrent users per second

1.5.1.1 The system MUST be capable of processing bulk searches and produce search reports for each data subject within the batch.